Mansha Ul Haq

in/manshauh github.com/manshaulhaq manshaulhaq2005@gmail.com Islamabad, PK +92 317 7985710

SUMMARY

AI Engineer building production-grade machine learning systems that bridge the gap between research and reality. Architects the full pipeline—curating data, training deep learning models (CNNs, Transformers, ensemble methods), and deploying low-latency APIs. Python, SQL, and a Linux-first workflow that prioritizes automation and efficiency.

WORK EXPERIENCE

Eziline Software House Jul 2025 – Sep 2025
Machine Learning Intern Rawalpindi, Pakistan

- Built and evaluated regression/classification models (Boston Housing, Titanic) using RMSE, R^2, and F1-score to ensure high predictive accuracy.
- Optimized model performance via Recursive Feature Elimination (RFE) and Hyperparameter Tuning using Grid and Randomized Search.
- Developed Deep Learning models in TensorFlow/Keras, including CNNs for image classification and LSTMs for text generation.
- Applied Transfer Learning by fine-tuning pre-trained ResNet50 models and freezing layers to optimize training for custom datasets.
- Performed customer segmentation using K-Means clustering, validating clusters with Elbow and Silhouette methods for data-driven insights.

PROJECTS

Real-Time Helmet Detection System | *Python, YOLOv8, OpenCV, Streamlit* [GitHub]

- Curated and augmented a Roboflow dataset (640×640, grayscale, brightness) for robust training.
- Trained a YOLOv8 Nano model over 50 epochs, validating with confusion matrices and precision-recall curves.
- Configured optimal 0.48 confidence threshold and deployed a Streamlit web app with live webcam inference.
- Achieved 77.1% mAP@0.5 and 86% compliance accuracy, minimizing missed helmet detections.

Social Media Sentiment Analysis | *Python, BERT, FastAPI, Hugging Face* [GitHub]

- Fine-tuned bert-base-uncased on cleaned social media data with lowercasing, URL/strip, and mention removal.
- Deployed a FastAPI REST API with asynchronous lifespan management for low-latency inference.
- Built a TF-IDF module to extract key drivers behind positive and negative sentiments.
- Delivered high-accuracy sentiment classification with real-time API access and automated recommendations.

Credit Risk & Loan Default Prediction | *Python, Scikit-learn, XGBoost, FastAPI* [GitHub]

- Built a complete data pipeline with missing value imputation, categorical encoding, and feature scaling.
- Trained an ensemble classifier (Random Forest/XGBoost) optimized for ROC-AUC and precision-recall curves.
- Serialized the model using joblib and deployed a REST API for real-time risk grading (A–D).
- Identified critical predictors (DTI ratio, credit utilization, loan purpose) for faster loan approvals.

EduBot | *Python, Gemini API* [GitHub]

- Built a user-friendly interface using Streamlit for seamless interaction.
- Designed to assist with academic and coding-related queries.
- Integrated media upload functionality to enhance query context.

EDUCATION

University of Engineering & Technology Sep 2023 – Jun 2027
Bachelor of Science in Computer Science
Coursework: Data Structures & Algorithms, OOP, DBMS, Artificial Intelligence, Computer Networks, Assembly Language

SKILLS

Languages: Python, R, C/C++
Libraries & Frameworks: TensorFlow, Keras, PyTorch, Scikit-Learn, Pandas, NumPy, Matplotlib
Databases: SQL/MySQL, MongoDB
OS & Tools: Arch Linux (daily driver), Git, Docker, FastAPI, Streamlit